

STAT 301: Statistical Modelling for Data Science

Statistical Designs and Causality

"*x causes y*" $\neq$ "*x and y are associated*"

Causal questions

regression:
find relationship
(association)

- Does *smoking* cause *cancer*?
- Do *lunch support programs* improve student test scores?
- Does *time spent on social media* make people *happier*?
- Does *driving while intoxicated* increase *accident rates*?
- Does *breastfeeding* increase baby *IQs*?

Reverse causality

Parental Involvement Is Overrated.

“When we examined whether regular help with homework had a positive impact on children’s academic performance, we were quite startled by what we found. Regardless of a family’s social class, racial or ethnic background, or a child’s grade level, consistent homework help almost never improved test scores or grades... Even more surprising to us was that when parents regularly helped with homework, kids usually performed worse.”

Likely children don’t performing well at school need and receive regular parental help

From <http://rafalab.dfci.harvard.edu/dsbook/association-is-not-causation.html>

Simpson's paradox

The sign of the **correlation flip** when comparing the entire publication and specific strata.

<https://simplystatistics.org/posts/2017-08-08-code-for-my-educational-gifs/>

UC Berkeley gender bias

In 1973, admission figures showed a statistically significant difference in the number of men admitted to the UC Berkeley compared to women.

However, when analyzing the data within departments, the different rejection percentages reveal a different trend. Women tended to apply to more competitive departments with lower rates of admission

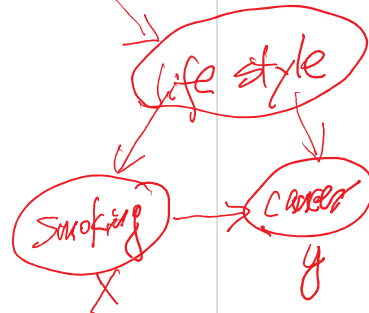
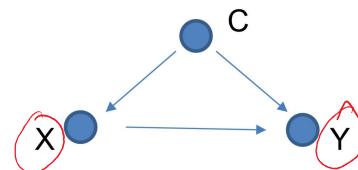
Confounding factors

A *confounder* is a variable that causes changes in both the response *and* at least one input variable.

Sports Analytics

A study of baseball data show that Home Runs was a confounder that resulted in a higher correlation than expected when studying the relationship between Bases on Balls and Runs.

From <http://rafalab.dfci.harvard.edu/dsbook/association-is-not-causation.html>



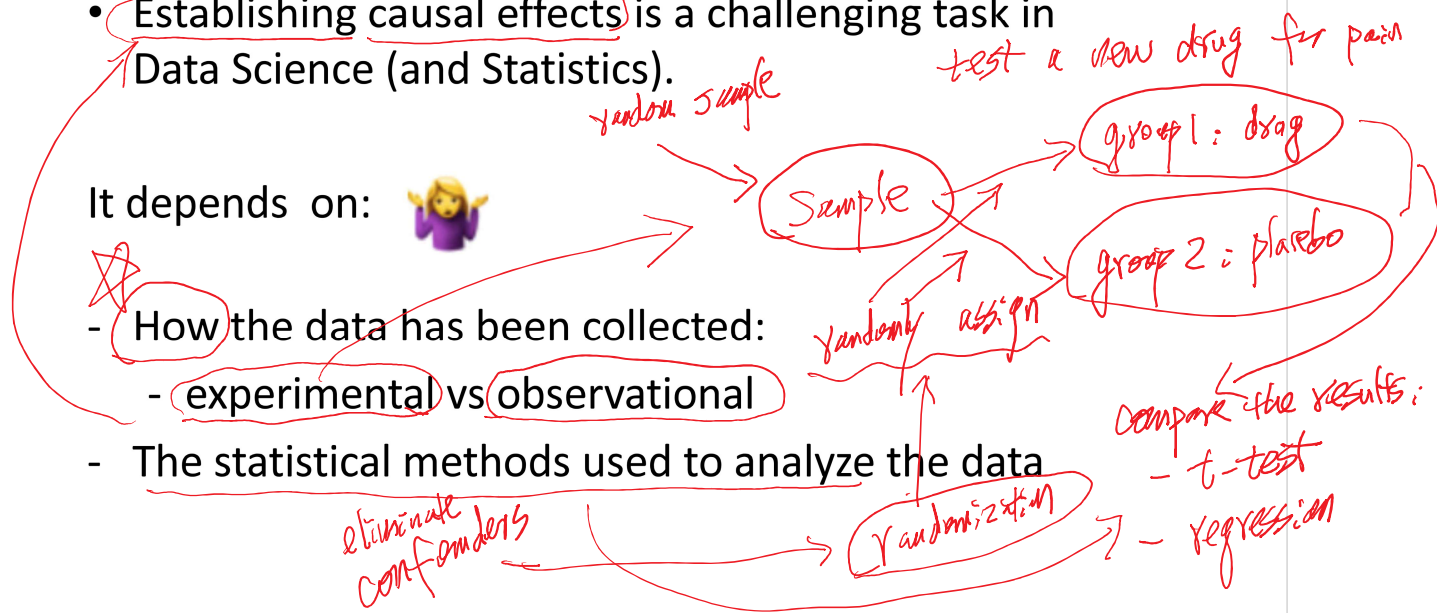
Causal Inference

- Establishing causal effects is a challenging task in Data Science (and Statistics).

It depends on:



- How the data has been collected:
 - experimental vs observational
- The statistical methods used to analyze the data



Experimental designs

- **Experiment design:** the manner in which the randomization of experimental units to treatments is carried out and how the data are collected. *key*

→ Simplest
Completely randomized design (CRD): experimental units are randomized throughout the data layout. No correlation between observations.

- Different pots were randomly assigned to different sulfur-nitrogen combinations

In a CRD, observed and unobserved confounders are balanced, on average. It is considered the gold standard design for causal inference. *→ eliminated*

Experimental designs (cont.)

smaller SD

Randomized Block design (RBD): splits experimental units into homogeneous blocks to remove variation from nuisance factors. Then, randomly assigns treatments to each block. Blocks are similar in all aspects except treatment!

In a RBD, only observed confounders are balanced so only average treatment effects can be estimated (using appropriate methods)

test stat

$$Z = \frac{\beta}{SE(\beta)}$$

smaller SE

$\Rightarrow$ smaller p-value

Observational study

Observational data: we collect data by measuring variables or surveying members without applying any treatment to them.

- We collect data from random sample of UBC students and examine their social media habits. Each person is classified as either a light, moderate, or heavy social media user.

In observational studies, treatments are not controlled by design! observed confounders can be included in the analysis but unobserved ones usually exist. Thus, causal effects *can not* be naively established.

smoking $\rightarrow$ cancer
Q: Can we do an experiment?

Simulation in tutorial 03

- Suppose a sports company's marketing team has developed a new video for their TikTok ad
- They want to know if this new ad will increase the customers' engagement (*ad dwell time* in seconds) compared to that of the *current ad*
- Suppose that athletes in this population have larger dwell times and are more interested in company innovations.

Simulation Design (dwell times in sec)

population means	current	new
→ non-athlete	15	23
→ athlete	20	28

The diagram illustrates a confounding relationship. A blue box labeled 'Athlete effect' has a blue arrow pointing to the 'new' column of the table. A red box labeled 'video effect' has a red arrow pointing to the 'current' column. A red handwritten note 'confounder' is written next to the 'Athlete effect' box. Additionally, red arrows point from the 'non-athlete' and 'athlete' labels to their respective rows in the table.

If athletes' tend to be more alert to updates in the sports companies. Thus, in an observational study, we may sample mostly athletes in the `new-ad` group and mostly non-athletes in the `current\_ad`.

If that occurs, the video effect can be confounded by the athlete effect.

Data generation

athlete	y_current	y_new	x_choice	y_observed	y_random
Y	$\mathcal{N}(20,1)$	$\mathcal{N}(28,1)$	new	$\mathcal{N}(28,1)$	$\mathcal{N}(20,1)$
Y	$\mathcal{N}(20,1)$	$\mathcal{N}(28,1)$	new	$\mathcal{N}(28,1)$	$\mathcal{N}(28,1)$
N	$\mathcal{N}(15,1)$	$\mathcal{N}(23,1)$	current	$\mathcal{N}(15,1)$	$\mathcal{N}(23,1)$
Y	$\mathcal{N}(20,1)$	$\mathcal{N}(28,1)$	new	$\mathcal{N}(28,1)$	$\mathcal{N}(20,1)$
N	$\mathcal{N}(15,1)$	$\mathcal{N}(23,1)$	current	$\mathcal{N}(15,1)$	$\mathcal{N}(15,1)$
...	...	...	...	...	...

Athlete
influences time
(Y), for both
type

Athlete
influences
choice of
type (X)

Thus 'athlete' is
a confounder

